ZNEXORA TECH | SECURITY BRIEFING

Windows Hardening



Protecting Enterprise Endpoints Through Secure Configuration

A Practical Guide for System Administrators and Security Teams

Presented by: **ZNexora Tech**

REFERENCES

- Microsoft Security Baselines
- CIS Benchmarks
- NIST Cybersecurity Framework

What is Windows Hardening?

Windows hardening is the process of implementing security controls and secure configurations to reduce vulnerabilities and limit opportunities for attackers.



OBJECTIVES

- Prevent unauthorized access
- Reduce cyber risks
- Strengthen endpoint defenses
- Improve security visibility
- Support regulatory compliance

Why Windows Hardening Matters



Credential Theft

Pass-the-Hash · Password Spraying · Credential Dumping

Malware & Ransomware

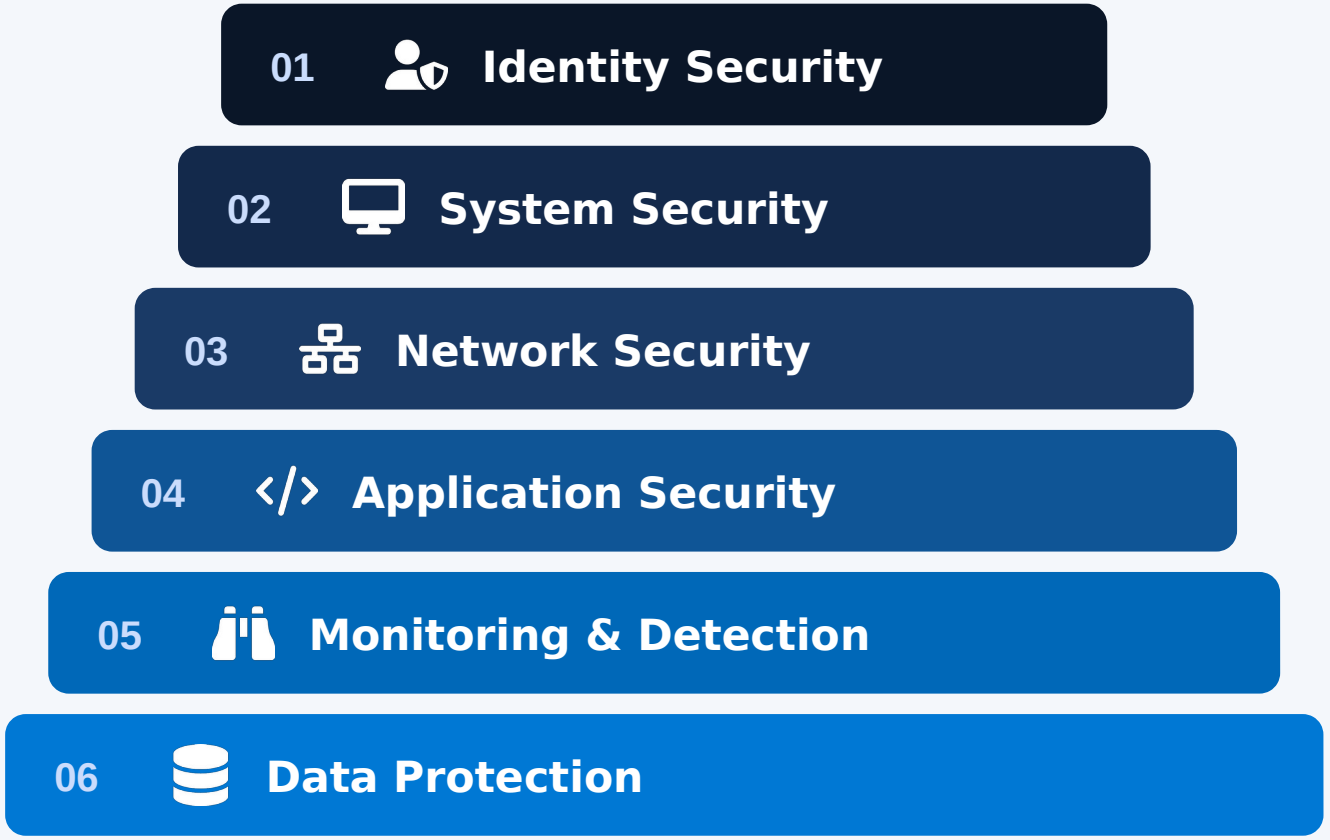
LockBit · Akira · Black Basta

Misconfigurations

Weak permissions · Open RDP · Legacy protocols

KEY MESSAGE Attackers often exploit weak configurations before exploiting zero-day vulnerabilities.

Windows Hardening Framework



GOAL

Secure every layer of the Windows environment

Independent controls are stacked, so if one layer fails, the next one still protects the endpoint.

Secure User & Administrator Accounts

RECOMMENDED CONTROLS

- Multi-Factor Authentication
- Least Privilege Principle
- Separate Admin Accounts
- Privileged Access Workstations
- Account Lockout Policies



BENEFITS

Reduced credential compromise

Reduced lateral movement

Improved access control

Patch & Vulnerability Management

1 Scan → 2 Prioritize → 3 Test → 4 Deploy → 5 Verify

CRITICAL AREAS

- **Windows Updates**
- **Microsoft Office**
- **Microsoft Edge**
- **Third-Party Applications**

BEST PRACTICES

- Monthly patch cycle
- Emergency patch deployment
- Vulnerability scanning
- Risk-based remediation

Microsoft Defender Security Stack



SECURITY BENEFITS

Real-time protection
Scans files and processes as they run

Behavioral detection
Flags suspicious activity, not just known malware

Threat intelligence
Cloud-delivered insight on emerging threats

Endpoint visibility
Central view of device health and alerts

Attack Surface Reduction



RECOMMENDED CONTROLS

- Block malicious macros
- Block Office child processes
- Restrict script execution
- Block credential theft attempts
- Control executable files

RESULT **Reduced ransomware and malware infections**

RDP & Network Hardening



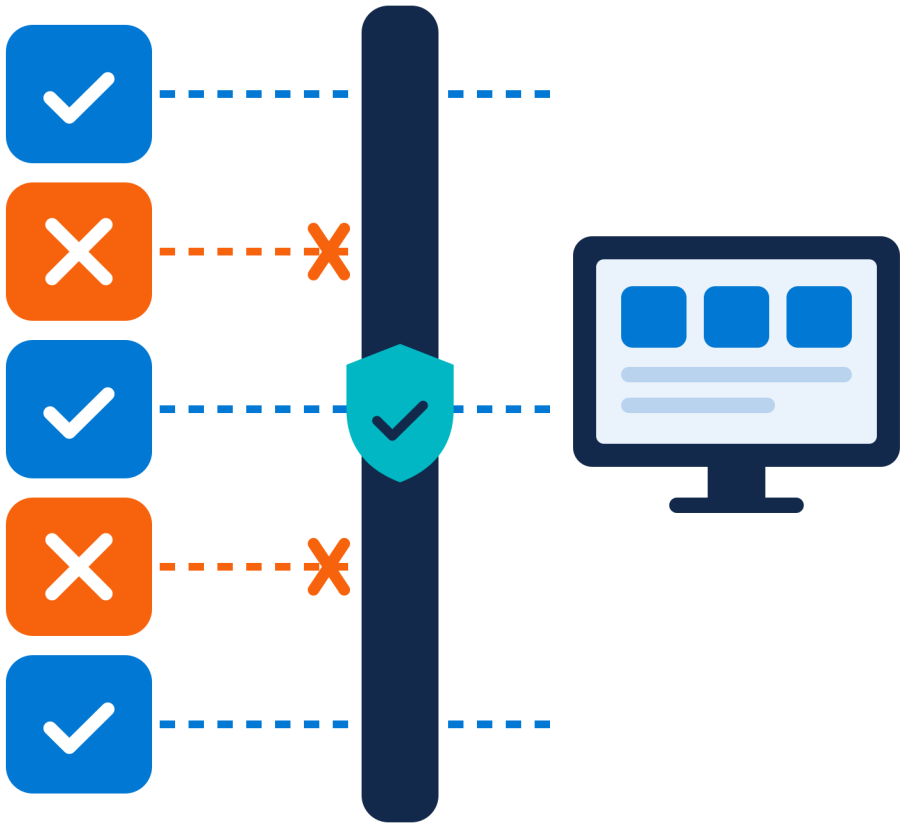
RDP HARDENING

- Network Level Authentication
- VPN Access Only
- MFA Enforcement
- Restricted Administrative Access

NETWORK HARDENING

- Windows Firewall
- SMBv1 Disabled
- TLS 1.2 / TLS 1.3
- Unused Services Disabled

Application Control



Microsoft AppLocker

Controls:

- Executables
- Scripts
- MSI Installers
- DLL Files

Windows Defender Application Control

Enforces trusted application execution.

- Reduced malware infection
- Reduced insider threats

Logging, Monitoring & Detection



MONITOR

- Account Logons
- Privilege Escalation
- Group Membership Changes
- PowerShell Activities
- Security Events

INTEGRATE WITH

Microsoft Sentinel | Azure Monitor | SIEM Platforms

Data Protection with BitLocker

ADVANTAGES

Full Disk Encryption
Protects all data on the drive

Lost Device Protection
A stolen laptop reveals nothing

Compliance Support
Helps meet data protection rules

Secure Remote Workforce
Safeguards devices off-site



RECOMMENDED

- TPM Enabled
- Recovery Key Management
- AES-256 Encryption

Windows Hardening Checklist

- **MFA Enabled**
- **Windows Fully Patched**
- **Defender Enabled**
- **Firewall Enabled**
- **BitLocker Enabled**
- **RDP Secured**
- **ASR Rules Applied**
- **AppLocker Configured**
- **Security Logs Monitored**
- **Backups Tested**

Review this baseline regularly: hardening is a continuous process, not a one-time project.

Key Takeaways

Protect Identities
MFA, least privilege, separate admin accounts

Secure Configurations
Baselines, ASR rules and application control

Patch Continuously
Monthly cycle plus emergency patching

Monitor Everything
Centralise logs in your SIEM

Encrypt Sensitive Data
BitLocker with managed recovery keys

Prepare for Incidents
Tested backups and response plans

FINAL MESSAGE

Secure Windows systems are built through continuous hardening, monitoring, and improvement.

QUESTIONS & DISCUSSION

Thank You

Windows Security Hardening: Practical Defense for Modern Enterprises

Presented by: **ZNexora Tech**



We welcome your questions and experiences.